

Consider the following grammar which is intended for recognition of a Bibtex file.

```
P → A NL P | B NL P | I NL P | A | B | I
A → @ARTICLE BLOCK1
B → @BOOK BLOCK2
I → @INPROCEEDING BLOCK3
BLOCK1 → {article-STEXT, NL
          TAB author = {TEXT}, NL
          TAB title = {ALPHANUMERIC}, NL
          TAB journal = {TEXT Journal}, NL
          TAB year = DIGIT, NL
          TAB pages = DIGIT, NL
          } NL
BLOCK2 → {book-STEXT, NL
          TAB author = {TEXT}, NL
          TAB title = {ALPHANUMERIC}, NL
          TAB volume = DIGIT, NL
          TAB publisher = {TEXT}, NL
          } NL
BLOCK3 → {inproceedings-STEXT, NL
          TAB author = {TEXT}, NL
          TAB title = {ALPHANUMERIC}, NL
          TAB editor = {TEXT}, NL
          TAB series = {TEXT}, NL
          TAB year = DIGIT, NL
          TAB publisher = { ALPHANUMERIC }, NL
          } NL
STEXT → full | minimal | crossref | journal
```

NOTE: some of the specified keywords are: @ARTICLE, @BOOK, @INPROCEEDING, article, title, journal, year, pages, volume, publisher, editor, series, full, minimal, crossref, journal, NL, TAB, {, }, =, ,(comma). NL represents newline.

TEXT represents a word (consisting of a sequence of alphabets only).

ALPHANUMERIC represents a string including space consisting of alphabets and digits.

DIGIT represents a sequence of digits.

1. Write the lex and yacc programs for this grammar. Your parser should be able to detect syntax errors and print an appropriate error message to indicate the position and nature of the error, including whether it is a lexical error or a grammatical error. You may terminate on detecting the first error.
2. Use this program to convert the corresponding bibtex file into xml file with appropriate tags as specified in the bibtex file.

Sample Bibtex file

```
@ARTICLE{article-full,  
author={Leslie},  
title={The Gnats and Gnus Document Preparation System},  
journal={Animal},  
year=1986,  
page=73,  
}  
@BOOK{book-full,  
author={Donald},  
title={Seminumerical Algorithms},  
volume=2,  
publisher={Addison Wesley},  
}  
@INPROCEEDINGS{inproceedings-minimal,  
author={Alfred},  
title={Lower Bounds on wishful research results},  
editor={Wizard},  
series={ACM},  
year=1990,  
publisher={Academic Press},  
}  
@BOOK{book-full,  
author={Donald},  
title={Seminumerical Algorithms},  
volume=2,  
publisher={Addison Wesley},  
}
```

Sample Output

```
<document>
<ARTICLE>
  <article>full</article>
  <author>Leslie</author>
  <title>The Gnats and Gnus Document Preparation System</title>
  <journal>Animal</journal>
  <year>1986</year>
  <page>73</page>
</ARTICLE>
<BOOK>
  <book>full</book>
  <author>Donald</author>
  <title>Seminumerical Algorithms</title>
  <volume>2</volume>
  <publisher>Addison Wesley</publisher>
</BOOK>
<INPROCEEDINGS>
  <inproceedings>minimal</inproceedings>
  <author>Alfred</author>
  <title>Lower Bounds on wishful research results</title>
  <editor>Wizard</editor>
  <series>ACM</series>
  <year>1990</year>
  <publisher>Academic Press</publisher>
</INPROCEEDINGS>
<BOOK>
  <book>full</book>
  <author>Donald</author>
  <title>Seminumerical Algorithms</title>
  <volume>2</volume>
  <publisher>Addison Wesley</publisher>
</BOOK>
</document>
```